Introduction and Perpetual growth through factor accumulation

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Introduction

Some practical information

- Taught by myself (Part I) and Asger Wingender (Part II)
- Reading list is found at Absalon page (subject to change)
  - The pdf containing reading list also contains the lecture plan which allows you to keep track of where we are if you have had to be absent.
- Weekly meetings: 3 hours
  - First hour: Exercises.
    - No printed solution manual; but either me or Asger is here to help ...
    - Reflect type of questions featured on the final.

Introduction

Some practical information

- Next two hours: Lecture
  - Usually part theory, part empirics.
  - I priortize the lecture. If some part of the readings are not explicitly discussed do not assume it is not relevant. It may mean that I think (erronously, perhaps) that you can easily read it for yourself. Ask!!
  - Largely papers/articles. They are available online free of charge via the University
- Office hours: Wednesdays 13-14. Note: Not for questions regarding exercises! Questions about required reading; discussion of thesis ideas etc.

Introduction

Some practical information

- Graduate course: Certain knowledge is assumed
  - Working knowledge of basic growth models (Solow-Swan, Diamond + Ramsey/Cass/Koopmans). B&S Ch. 1-3
    - Optimal control theory is used at times but not assumed known ("cookbook" approach – see also B&S Appendix)
  - Working knowledge of econometrics (Cross-section; Panel regressions etc)
- Final
  - Tested in being able to solve models and elicit policy recommendations and/or empirical implications
  - Partly analytical (about 60%), partly "verbal". Exercises will provide examples.

Introduction

Some practical information

- Related master level courses?
  - Advanced development economics – macro
  - Environmental- Ressource and climate economics
  - No "money" in present course– classical dichtonomy maintained. So Advanced Monetary Economics is a good supplement.
  - "Input" to many other course including labor and finance
- Questions?

Roadmap

1. Some practical information
2. **Perpetual growth through factor accumulation**
   1. **Technological requirements**
   2. The role of preferences
3. Endogenous savings: Requirements for Endogenous Growth
   1. Is perpetual growth optimal?
   2. Policy?
4. Remarks on empirical evidence
5. Learning objectives

Perpetual growth through factor accumulation

- To words about terminology: A common distinction between growth models, where the growth rate is determined within the model, is between "factor-based" growth models and "idea-based" growth models
  - Factor-based is models where long-run growth is driven by standard inputs: physical capital, human capital.
  - Idea-based growth models refer to settings where innovation is endogenous. These theories are somewhat more involved as they require a departure from competitive markets. In a coming lecture we will discuss why this is inevitable.
- Today: Factor-based growth models. These models are all reconsilable with competitive markets, which makes them simpler to work with of course. Perfect competition is therefore assumed throughout this lecture.

Perpetual growth through factor accumulation

- Throughout this course we will at times work with models where growth is determined *within* the model: "Endogenous" growth.
- Useful to have a working definition of what we mean when we say:

Definition

Endogenous growth is said to be present if perpetual growth is feasible without variables in the model are growing at exogenous rates.

- Notice: "Feasible". Points to technological requirements. Preferences will be decisive when it comes to "viability".

Perpetual growth through factor accumulation

Technological requirements

- To focus on technological requirements, household optimization is surpressed.
- Consider a standard Solow-Swan model, with no technological change and no population growth.
- Time is continuous (index supressed). Output $y = f(k)$; s is the savings rate and δ the rate of depreciation.
- Law of motion:

$$\frac{\dot{k}}{k} = s\frac{f(k)}{k} - \delta.$$

- In the limit when k becomes "big":

$$\lim_{k\to\infty}\frac{\dot{k}}{k} = s\lim_{k\to\infty}\frac{f(k)}{k} - \delta \overset{\text{L'Hopital}}{=} s\lim_{k\to\infty} f'(k) - \delta$$

- For perpetual growth we thus cannot allow the standard Inada condition, $\lim_{k\to\infty} f'(k) = 0$. Output needs to be unbounded.

Perpetual growth through factor accumulation

Technological requirements

- Examples that do away with the Inada condition:

1. The "AK model": $y = f(k) \equiv Ak$. Obviously, $f'(k) = A > 0$, so the Inada condition is violated.
2. The *asymptotic* AK model (Jones and Manuelli, 1990): $\lim_{k\to\infty} f'(k) = A > 0$.

- In either case:

$$\lim_{k\to\infty} \frac{\dot{k}}{k} = s \lim_{k\to\infty} f'(k) - \delta = sA - \delta.$$

- Perpetual growth is *feasible* even absent variables growing at exogenous rates: endogenous growth (cf definition).
- *The requirement for endogenous growth is that the marginal product of capital is bounded away from zero.*

Perpetual growth through factor accumulation

Technological requirements

- An *equivalent* way of stating the requirement:
  - *Endogenous growth requires (asymptotically) constant returns to scale in reproducible factors of production*.

1. AK model: $y = Ak$. One reproducible factor of production: k. Requirement: $\frac{\partial y}{\partial k}\frac{k}{y} = 1$. Check: $\frac{\partial y}{\partial k} = A$, $\frac{\partial y}{\partial k}\frac{k}{y} = A\frac{k}{Ak} = 1$.
2. Asymptotic AK model ($\lim_{k\to\infty} f'(k) = A > 0$). Check: $\frac{\partial y}{\partial k}\frac{k}{y} = \frac{f'(k)k}{f(k)} \neq 1$. But in the limit

$$\lim_{k\to\infty} \frac{f'(k)\,k}{f(k)} = A \lim_{k\to\infty} \frac{k}{f(k)} = A \lim_{k\to\infty} \frac{1}{f'(k)} = 1.$$

Perpetual growth through factor accumulation

Technological requirements

- Hence, two equivalent ways of stating the requirement for endogenous growth:

1. *The marginal product of capital is bounded away from zero (asymptotically)*

$$\lim_{k \to \infty} f'(k) = A > 0$$

2. *Constant returns to reproducible factor(s) of production (asymptotically)*

$$\lim_{k \to \infty} \frac{\partial y}{\partial k} \frac{k}{y} = 1$$

- If this requirement is fulfilled perpetual growth is *feasible*, *regardless* of market structure (competitive, monopolistic etc).

Perpetual growth through factor accumulation

Mechanisms: Learning-by-doing

- There is a number of ways in which the AK structure can be *derived* (rather than simply assumed, as in the examples above). B&S Ch. 4.
- **Externalities**: "Learning-by-doing"/Learning-by-investing
  - Through productive experience you gain knowlegde as a *biproduct.*
  - Important: *External* learning. "My productive experience ultimately benefit *other* producers." (example: "Liberty shipyard "miracle" during WWII).
  - Modelling. The representative firm, acting in competitive markets, face the technology

$$Y(t) = K(t)^{\alpha} [A(t) L]^{1-\alpha}$$

As usual $A(t)$ is taken as given by the individual firm. New part: Assume $A(t) = AK(t)$. External learning-by-investing.

Perpetual growth through factor accumulation

Mechanisms: Learning-by-doing

- When you take the externality into account

$$Y(t) = K(t)^{\alpha} \left[AK(t) L\right]^{1-\alpha} = A^{1-\alpha} L^{1-\alpha} K(t),$$

  which exhibits constant returns to K (the only reproducible factor) $\Rightarrow$ requirement for endogenous growth is fulfilled.
- Is this is good explanation for perpetual growth? Not in itself, unfortunately. To see why, notice I made a strong assumption $A(t) = AK(t)^{\lambda}$, with λ exactly equal to 1. If not, we have either increasing ($\lambda > 1$), or diminishing returns to K.What does the data say?

Perpetual growth through factor accumulation

Mechanisms: Learning-by-doing

- Thornton and Thompson (2001) examines production of ships during WWII
  - Same ships (including the "Liberty ship") produced throughout the war $\Rightarrow$ lots of room for learning
  - Multiple yards allows you to ask whether production had benefits across ship yards producting the same ship and/or other ships. *External* learning.
  - Maximum chance for spillovers: unified effort viz the Nazi's; deliberate sharing of knowlege etc
  - TT do find evidence of external learning. But not to the extent required for endogenous growth ($\lambda = 1$). See also Irwin and Klenow (1994) on semi-conductors and learning.

Perpetual growth through factor accumulation

Mechanisms: Government investments

- **Government Investments.** Governments have the power to make investments that potentially increase productivity
  - E.g. infrastructure; Public R&D etc.
- How would that work? The representative firm, acting in competitive markets, face the technology

$$Y(t) = K(t)^{\alpha} (A(t) L)^{1-\alpha}$$

As usual $A(t)$ is taken as given by the individual firm. New part: Assume $A(t) = AG(t)$, where G is government investments (say in infrastructure). Note that if the government operates under a balanced budget

$$G(t) = \tau Y(t),$$

where I assume a flat tax on income or production (and a time constant tax rate in this example).

Perpetual growth through factor accumulation

Mechanisms: Government investments

- Putting the elements together we have the following:

$$Y(t) = K(t)^{\alpha} [G(t) L]^{1-\alpha} = K(t)^{\alpha} [\tau Y(t) L]^{1-\alpha}$$

- Isolating for Y:

$$Y(t) = [\tau L]^{\frac{1-\alpha}{\alpha}} K(t)$$

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Perpetual growth through factor accumulation

Mechanisms: Government investments

- In the wake of Ashauer (1989) a sizable literature has explored infrastructure investments. See also Fernald (1999).
  - Significant effects on productivity
  - Not possible to account for $\lambda = 1$ requirements.
- Obviously, other sources include public investments in e.g. R&D. So far, however, no evidence suggest that we can provide evidence of a "pure G" story of endogenous growth
  - On the other hand it is not easy a priori to refute that $\lambda = 1$ could arise due to a number of different channels; externalities, government investments etc. Strict requirement, admittedly, but in the end an empirical question.

Perpetual growth through factor accumulation

Mechanisms: "Broad" capital

- **Broad capital.** Finally, a third option is to add more accumulated factors:

$$Y = AK^{\alpha}H^{1-\alpha}$$

- Suppose symmetrical production of K and H (say, constant savings rate out of output builds up K and H, like the Mankiw-Romer-Weil model: $\dot{H} = s_H Y$, $\dot{K} = s_K Y$, staying with exogenous savings). Again: Constant returns to accumulated factor**s** of production (H,K) $\Rightarrow$ endogenous growth arises.
- Perhaps elegant from technical perspective, but surely example of "mathiness" (Romer, 2015).
  - Constructing K and H is *not* the same thing, and thus would require *different* production functions. One good setting: *same* technology. Multisector setting: the CRTS requirement is still sufficient (in sector producing "capital"), but not nessesary in all sectors; see discussion in B&S Ch. 5.

Perpetual growth through factor accumulation

Summing up

- The key assumption needed for endogenous growth: MPK needs to be bounded away from zero (or, equivalently, constant returns to scale in reproducible factor).
  - Leads to the "AK-model" (at least when k becomes very large)
- Several different ways to this structure: Multiple investment goods (i.e., "broad capital"); Externalities (e.g., Learning-by-doing); Productive government investments.
- Since there are different ways to "get to AK", one can think of AK-models as a "class of models", reflecting everything from "pure AK" to productive government investments.

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Perpetual growth through factor accumulation

The role of preferences

- So far we have focused on technological requirements. But whether growth in the end arises or not also depends on preferences.
- To see this in a simple way, notice that in the Solow-Swan setting the "asymptotic AK model" implies (here I set $\delta = 0$ to simplify)

$$\lim_{k \to \infty} \frac{\dot{k}}{k} = s \lim_{k \to \infty} \frac{f(k)}{k} = s \lim_{k \to \infty} f'(k) = sA$$

- Consider, next, a different assumption about savings (in this sense, "preferences"). Generalizing the Solow-Swan assumption slightly

$$s = s^r rk + s^w w$$

where the standard Solow-Swan assumption would be $s^r = s^r = s$ (recall, under competitive markets, $y = rk + w$).

Perpetual growth through factor accumulation

The role of preferences

- With competitive markets, we have the following from the representative firms first order conditions (time is suppressed):

$$w = f(k) - kf'(k)$$

$$r = f'(k)$$

- Law of motion for capital

$$\frac{\dot{k}}{k} = s^r f'(k) + s^w \left(\frac{f(k)}{k} - f'(k) \right)$$

- A particular stark result obtains if $s^r = 0$, $s^w > 0$

$$\lim_{k \to \infty} \frac{\dot{k}}{k} = s^w \lim_{k \to \infty} \frac{f(k)}{k} - s^w \lim_{k \to \infty} f'(k) = 0.$$

- So if $s = sy$ (Solow-Swan) the AK-structure gives us perpetual growth. If $s = sw$ it does not.

Perpetual growth through factor accumulation

The role of preferences

- Hence, whether perpetual growth is *viable* is highly dependent on preferences.
- Why would $s^r = 0$, $s^w > 0$ arise? It would be the outcome within a two-period OLG model (Jones and Manuelli, 1992, on the reading list, is making this point)
- In contrast, within the Ramsey-Cass-Koopmans framework consumers save a constant fraction of capital income along a steady state path. Hence, under similar *technological* assumptions you get radically different outcomes.
- At a deeper level it comes down to the savings *motive*: Life-cycle motive (OLG), pure bequest motive (Ramsey-Cass-Koopmans). (See discussion in B&S p. 198-200).

Perpetual growth through factor accumulation

The role of preferences

- Hard to over-emphasize the importance of savings motive for growth
- As you know, with pure life-cycle motive it is possible to see: Dynamic inefficiency; multiple equilibria (temporary policy may have permanent effects!); zero impact on prosperity from capital taxation (but wage taxation matters to long-run prosperity).
- Pure bequest motive: There is never dynamic inefficiency; There is always a unique steady state; capital taxation is bad (but wage taxation doesn't matter) for long-run prosperity.
- To this list we can now add differences in predictions wrt the viability of perpetual growth (Asympotic AK model does not allow for growth in the Life-cycle setting).
- The same is true for another vexing prediction: *Scale effects*.

Perpetual growth through factor accumulation

The role of preferences

- To see this clearly, consider the **learning-by-doing framework (Cf B&S Ch. 4)**: $Y = F(K, AL) = f\left(\frac{AL}{K}\right)K$, with $A = BK$. **Note**: $f' > 0$, $f'' < 0$. First order conditions from the representative firm problem (again, $\delta = 0$):

$$\frac{\partial Y}{\partial K} = f\left(\frac{AL}{K}\right) - f'\left(\frac{AL}{K}\right)\left(\frac{AL}{K_i}\right) = r$$

$$\frac{\partial Y}{\partial L} = f'\left(\frac{AL}{K}\right)A = w$$

- In equilibrium (using $A = BK$), factor prices fullfill $r = f(BL) - f'(BL)(BL)$, $w = f'(BL)BK$
- Suppose, first, we have the standard Solow-Swan form

$$\dot{k}/k = sy/k = sf(BL).$$

As $f' > 0$ it follows that a bigger population leads to faster growth – the scale effect. **Hard to reconsile with data.**

Perpetual growth through factor accumulation

The role of preferences

- Suppose now that savings are *a la* life-cycle setting (savings from wage income)

$$\dot{k}/k = s^w w/k = sf'(BL)B.$$

- Since $f'' < 0$ it follows that a bigger population leads to *slower* growth! (if both motives – life cycle and bequest - it may leave the growth rate *unaffected*, Dalgaard and Jensen (2009))
- All of these predictions flows from the fact that the savings motive ultimately determines how savings react to changes in the *factor income distribution*.
- Hence, from a *positive* perspective perpetual growth from factor accumulation rests not only on technological requirements, but on a fuller understanding of the economy – notably, how people are motivated to save.
- Here we have thought of "preferences" loosely. Precludes welfare considerations explicitly. Let's remedy that...

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Is perpetual growth optimal?

- Naturally, we now have to specify preferences explicitly. One standard approach is the "Dynastic consumer".
- Consider the following social planner problem familiar from the standard Ramsey-Cass-Koopmans model. Closed economy. No population growth for simplicity

$$\max_{\{c\}_{t=0}^{\infty}} U_0 = \int_{t=0}^{\infty} \frac{c^{1-\theta}-1}{1-\theta} L e^{-\rho t} dt$$

$$c_t \geqq 0$$

$$\dot{K} = Y - C - \delta K, \ K_0 \text{ given}, \ C = cL$$

$$Y = F(K, L)$$

$$K_t \geqq 0 \text{ for all } t$$

(NB: non-negative capital rather than NPG since Social Planner; no sense in which capital can be "negative".)

Is perpetual growth optimal?

- The problem we face can be stated more generally

$$\max_{u(t)} \int_0^T e^{-\rho t} f(t, x, u)\, dt$$

$$s.t.\ \dot{x} = g(t, x, u)$$

$$x(0) = x_0$$

- Where x ("state variable") is a continuos variable, while u ("control variable") can move discontinuously and affects the state variable. NB: $T \to \infty$ is allowed.

Is perpetual growth optimal?

- Can be solved using **optimal control theory**. Define the "Hamiltonian":

$$J = e^{-\rho t} f(t, x, u) + \lambda(t) g(t, x, u)$$

- Nessesary conditions for optimal solution

$$\frac{\partial J}{\partial u} = 0$$

$$\frac{\partial J}{\partial x} = -\dot{\lambda}(t)$$

$$\frac{\partial J}{\partial \lambda} = \dot{x}$$

and

$$\lambda(T) x(T) = 0$$

Is perpetual growth optimal?

- Applied. Hamiltonian:

$$J = \frac{c^{1-\theta} - 1}{1-\theta} L e^{-\rho t} + \lambda(t)\left[F(K, L) - C - \delta K\right]$$

- Ness conditions:

$$\frac{\partial J}{\partial c} = 0 \Rightarrow c^{-\theta} L e^{-\rho t} = \lambda(t)$$

$$\frac{\partial J}{\partial K} = -\dot{\lambda}(t) \Rightarrow \lambda(t)\left(F'_K - \delta\right) = -\dot{\lambda}(t)$$

plus the transversality condition

$$\lim_{t \to \infty} \lambda(t) K(t) = 0.$$

and the flow constraint (from $\partial J / \partial \lambda = \dot{K}$)

Is perpetual growth optimal?

- **Optimal consumption** is derived in 4 steps: 1. log differentiate $\partial J/\partial c$

$$-\theta\frac{\dot{c}}{c}-\rho=\frac{\dot{\lambda}(t)}{\lambda(t)}$$

- 2: rearrange $\partial J/\partial K$

$$F_K'-\delta=-\frac{\dot{\lambda}(t)}{\lambda(t)}.$$

- 3: substitute from 1 into 2

$$\theta\frac{\dot{c}}{c}+\rho=F_K'-\delta$$

- 4: isolate for $\dot{c}/c$

$$\frac{\dot{c}(t)}{c(t)}=\frac{1}{\theta}\left[F_K'-\rho-\delta\right].$$

Is perpetual growth optimal?

- So, familiar Consumption Euler (or "Keynes-Ramsey rule")

$$\frac{\dot{c}}{c} = \frac{1}{\theta}\left(F_K' - \delta - \rho\right)$$

  (MRS = MRT at alle points in time)
- Observe that if F_K' is bounded away from zero, then perpetual growth in consumption is viable. But what about growth in Y?
- **Two useful results, when the production function can be written** $Y = F(K, L) = AK$ (in reduced form).
  - No transitional dynamics (Balanced growth)
  - We can infer the growth rate of Y from the Consumption Euler along an optimal path.

Is perpetual growth optimal?

- The **intuition** runs as follows. First, since $Y = AK$, we have that $\dot{Y}/Y = \dot{K}/K$ at all points in time.
- Second, observe that $\dot{C}/C$ is growing at a constant rate (since $F_k' = A$; cf consumption euler).
- Third, note that if $\dot{Y}/Y \neq \dot{C}/C$ then the savings rate $\left(1 - \frac{C}{Y}\right)$ tends either to zero or one
  - Neither can - intuitively - be optimal; violates either consumption euler or Transversality condition (Formal proof: B&S p.208-10)
  - As a result, only viable - optimal - solution, is $\dot{C}/C = \dot{Y}/Y = \dot{K}/K$ at all points in time. So: (i) growth rate evaluated from the consumption Euler gives you the growth rate of output (per capita); (ii) Balanced growth at all points in time. Holds in **all** one sector models where $Y = AK$ holds.

Is perpetual growth optimal?

- We need to check the solution is *optimal* in the presence of perpetual growth
- The *transversality condition* must be fulfilled:

$$\lim_{t\to\infty} \lambda_t K_t = 0$$

  In the AK model above $\lambda_t = \lambda_0 e^{-(A-\delta-\rho)t}$, and $K_t = K_0 e^{\gamma t}$. So $\lambda_t K_t \propto e^{-(A-\delta-\rho-\gamma)t}$. We need $A - \delta - \rho - \gamma > 0$, which is a matter of parameter values.
- *Discounted utility needs to be bounded* (otherwise optimization meaningless). Again in AK case

$$U_0 = \int_0^\infty \frac{c^{1-\theta}-1}{1-\theta} e^{-\rho t} L dt = \int_0^\infty \frac{c_0^{1-\theta} e^{(1-\theta)\gamma t}-1}{1-\theta} e^{-\rho t} L dt.$$

- We assume (unless compelled in some way to do otherwise) that parameters are such that growth is (positive and) optimal. Outside this course you should always check.

Policy?

- Hence, in the present case (Dynastic or "Ramsey" consumers) growth is optimal (subject to the caveat about parametervalues)
- Note, that is need not always be the case.
  - The OLG case is an exampe, where growth is not nessesarily optimal (cf Jones and Manuelli)
  - Later in the course (Asger's part) you will see another example of how it may not be
- Assuming positive growth is optimal. Should the government take actions that *raise* the growth rate above that generated by the market economy? Again, the answer is not clear-cut, as we will see next.

Policy?

- The traditional approach to deriving policy implications in macro
  - Examine the decentralized solution and the social planner solution
  - In very many applications (as above): Social planner maximizes the utility of the representative agent, subject to resource constraints (*implicitly assuming, therefore, the existence of a positive and normative representative agent*!)
  - Choose a policy such that the planner solution obtains in the decentralized setting
- In models without market failures – *cf "pure AK" model or "broad capital" approach* – the decentralized (market) solution and planner solution coinside (just as in the Ramsey-Cass-Koopmans model)
  - No role for welfare improving policy.

Policy?

- In most other models (Learning-by-Doing; Productive Government Investments) things are different
  - Learning-by-Doing. Firms do not internalize the positive externality from investing $\Rightarrow$ Too little investments from a societal point of view. $\Rightarrow$ subsidies (financed by non-distorting taxes; Lump-sum taxes, say). In other words: The market economy is growing to slow.
  - Productive government investments: Pure public good (cf above, $Y = K^{\alpha} (GL)^{1-\alpha}$). Choose G such $\frac{\partial Y}{\partial G} = 1$. Finance via lump sum taxes.
  - Productive government investments: Congestion (e.g., $Y = f\left(\frac{G}{Y}\right) K$... with balanced budget the reduced form is $Y = f(\tau) K$, thus "AK"). Choose G such $\frac{\partial Y}{\partial G} = 1$. Finance via distorting taxes! In other words: The market is growing too fast! (we will see another example where this is the case, later in the course, when innovation is endogenous).

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Remarks on empirical evidence

- Robust implication of the entire class of model (i.e., AK-models): Higher savings/investment rate leads to a permanently higher *growth rate.*
  - Contrast: Solow-Swan/ Ramsey-Cass-Koopmans suggest levels effects. That is *temporary* growth effects.
  - Note I. Many AK-models also suggest Scale effects, but, as we saw, this is NOT a robust implication as it depends on how savings are formed – bequest, life-cycle etc
  - Note II: Many AK-models have no transitional dynamics. Not a robust implication as the "asympotic AK-model" allows for transitional dynamics.
- A famous paper by Jones (1995) argues "$s \nrightarrow g$". More recently, revised by Bond et al (see reading list).

Remarks on empirical evidence

- The "meat and potatoes" of the paper concerns the empirical specification.
- Consider first a Neoclassical world where production is Cobb-Douglas, $Y_t/L_t = A_t\left(K_t/L_t\right)^{\alpha}$, and $k_t = s_{K,t}y_{t-1}$, (so ignoring depreciation).
- Taking logs and substiting law of motion for k into production function

$$y_t = a_t + \alpha\sigma_{K,t} + \alpha y_{t-1}$$

  where $\sigma_{K,t} \equiv \ln s_{K,t}$, $y_t = \ln\left(Y_t/L_t\right)$, $a_t \equiv \ln A_t$. This is nothing but the standard "convergence" equation.
- Take first differences:

$$\Delta y_t = \Delta a + \alpha\Delta\sigma_{K,t} + \alpha\Delta y_{t-1} \tag{1}$$

Remarks on empirical evidence

- If we are in a Neoclassical world the rate of technological change is exogenous ($a_t = a_0 + gt$):

$$\Delta a = g. \tag{2}$$

- A steady state of the model would require $\Delta y_t = \Delta y_{t-1} = \Delta y^*$ and $\Delta \sigma_K = 0$. We have (inserting in (1))

$$\Delta y^* = \frac{g}{1-\alpha}.$$

- Clearly investments do not affect growth (as its driven by exogenous tech change at rate g). Hence if we inspired by (1) run the regression

$$\Delta y_{it} = \alpha_i + \theta_t + \beta_1 \Delta \sigma_{K,t} + \beta_2 \Delta y_{t-1} + \varepsilon_{it} \qquad \text{(Neoclassic)}$$

significance of β_1 means investment rates only affect *levels.*

Remarks on empirical evidence

- Now, suppose we change equation (2) to

$$\Delta a = g + \phi\sigma_K$$

- Key implication of AK-type models: investments affect the growth rate of A. (see slides above for Learning-by-doing, and note that $A(t) = AK(t)$).
- This would suggest

$$\Delta y_t = g + \phi\sigma_{Kt} + \alpha\Delta\sigma_{K,t} + \alpha\Delta y_{t-1}$$

- In a steady state ($\Delta y_t = \Delta y_{t-1} = \Delta y^*$ and $\Delta\sigma_K = 0$)

$$\Delta y^* = \frac{g + \phi\sigma_K}{1 - \alpha}$$

- So where does that leave us?

Remarks on empirical evidence

- The key thing. We need the following empirical specification

$$\Delta y_{it} = \alpha_i + \theta_t + \beta_1 \sigma_{K,t} + \beta_2 \Delta\sigma_{K,t} + \beta_3 \Delta y_{t-1} + \varepsilon_{it}, \qquad \text{(AK)}$$

  where *both* the level and log change in σ_K enters; difference from equation (Neoclassic). To highlight the difference more starkly, note that $\Delta\sigma_{K,t} \equiv \sigma_{K_{,t}} - \sigma_{K,t-1}$. Substitute and rearrange in equation (AK)

$$\Delta y_{it} = \alpha_i + \theta_t + (\beta_1 + \beta_2)\,\sigma_{K,t} - \beta_2 \sigma_{K,t-1} + \beta_3 \Delta y_{t-1} + \varepsilon_{it}$$

  Hence "current period" investment rates and last period investment rates should enter with opposite signs! Significance of β_1 means that investment rates affect the growth *rate.*
- Bond et al find the data favors growth effects of $\sigma_{K,t}$, which supports AK-type models. Mostly, however, outside OECD. More work is needed, but AK models do seem to capture something about the growth process that Neoclassical models do not.

Learning objectives

1. You should understand the technical requirements for endogenous growth (i.e., so that you can check in a practical case whether the requirements are met)
2. You should be able to discuss the importance of *preferences* for the viability of growth (e.g., the no-growth result in Jones and Manuelli)
3. You should be able to work out the models discussed , including policy implications.
4. You should be able to solve the models in B&S Ch. 4. Both in the decentralized setting and the planner problem.
   - This means: being able to solve for the Consumption Euler (Keynes-Ramsey rule); explain - as well as show - why output growth and consumption growth coinside along an optimal path in AK models (i.e., no transitional dynamics result – balanced growth at all points in time). You should also be able to check that the derived growth rate is indeed optimal (satisfying the transversality condition etc).
5. You need to know the testing strategy of Bond et al and their overall finding.

Literature (not on reading list)

Aschauer, D. A. (1989). Is public expenditure productive?. Journal of monetary economics, 23(2), 177-200.

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Jones, L. E., & Manuelli, R. (1990). A convex model of equilibrium growth: Theory and policy implications. Journal of political Economy, 98(5, Part 1), 1008-1038.

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